

ORIGINAL RESEARCH

Disease-modifying interactions between chronic kidney disease and osteoarthritis: a new comorbid mouse model

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ABSTRACT

Objective The prevalence of comorbid chronic kidney disease (CKD) and osteoarthritis (OA) is increasing globally. While sharing common risk factors, the mechanism and consequences of concurrent CKD-OA are unclear. The aims of the study were to develop a preclinical comorbid model, and to investigate the disease-modifying interactions.

Methods Seventy (70) male 8–10 week-old C57BL/6 mice were subjected to 5/6 nephrectomy (5/6Nx)±destabilisation of medial meniscus (DMM) or sham surgery. OA pathology and CKD were assessed 12 weeks postinduction by blinded histology scoring, micro-CT, immunohistochemistry for osteoclast and matrix metalloproteinase (MMP)-13 activity, and serum analysis of bone metabolic markers.

Results The 5/6Nx model recapitulated characteristic features of CKD, with renal fibrosis and deranged serum alkaline phosphatase, calcium and phosphate. There was no histological evidence of cartilage pathology induced by 5/6Nx alone, however, synovial MMP-13 expression and subchondral bone osteoclastic activity were increased ($p<0.05$), with accompanying reductions ($p<0.05$) in subchondral trabecular bone, bone volume and mineral density. DMM significantly ($p<0.05$) increased tibiofemoral cartilage damage, subchondral bone sclerosis, marginal osteophytes and synovitis, in association with increased cartilage and synovial MMP-13. DMM alone induced ($p<0.05$) renal fibrosis, proteinuria and increased ($p<0.05$) 5/6Nx-induced serum urea. However, DMM in 5/6Nx-mice resulted in significantly reduced ($p<0.05$) cartilage pathology and marginal osteophyte development, in association with reduced subchondral bone volume and density, and inhibition of 5/6Nx-induced subchondral bone osteoclast activation.

Conclusion This study assessed a world-first preclinical comorbid CKD-OA model. Our findings demonstrate significant bidirectional disease-modifying interaction between CKD and OA.

INTRODUCTION

Osteoarthritis (OA) is the most common joint disease and leading cause of pain, disability and early retirement.¹ It is a complex disease affecting the entire joint,¹ multifactorial in

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The prevalence of comorbid chronic kidney disease (CKD) and osteoarthritis (OA) is increasing, but the mechanisms underpinning the association between CKD and OA are not known.

WHAT THIS STUDY ADDS

⇒ There is no preclinical model described in the literature to evaluate the molecular mechanisms of OA pathology associated with CKD. This study, for the first time, established a novel preclinical CKD-OA model and demonstrated significant bidirectional disease-modifying interactions between CKD and OA, supporting the observations with human patients.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This new preclinical model offers an exciting opportunity to explore the molecular pathways involved in OA-CKD cross-talk.

origin^{1,2} with its progression and impact worsened by the presence of other chronic health conditions.^{3–6} The prevalence of comorbid chronic kidney disease (CKD) and OA is increasing on a global scale.^{3–6} The reasons for this are not known but may be due to shared risk factors such as ageing, obesity, type 2 diabetes and hypertension. A large temporal cohort study in the UK indicated a bidirectional risk between incident OA and CKD,⁵ which could suggest disease-modifying biological cross-talk. OA is a frequent feature of the dialysis-dependent CKD population,⁷ demonstrating increased prevalence with duration of dialysis.⁸ Growth plates in children with CKD show abnormal chondrocyte maturation and delayed cartilage turnover,⁹ and knee cartilage thickness measured by ultrasound is reduced in patients with CKD.¹⁰ While these studies suggest CKD-induced

chondropathy may be implicated in increased OA risk, others have shown no change in radiographic knee or hip joint space width as a measure of cartilage loss in patients with CKD.¹¹

Renal osteodystrophy is common in patients with CKD and represents a complex disorder driven by concurrent biochemical and endocrine derangement.¹² Renal osteodystrophy affects bone quality and strength, and alterations in bone and mineral metabolism are part of a broader systemic CKD-Mineral and Bone Disorder (CKD-MBD).¹³ Subchondral bone provides mechanical support for cartilage, absorbing and distributing mechanical forces,² and CKD-MBD could, therefore, play a role in OA risk. While increased subchondral bone thickness and volume ('sclerosis') is a pathognomonic feature of late-stage OA that may increase mechanical loading and degradation of overlying cartilage, abnormal subchondral bone resorption and increased bone turnover in earlier disease stages is implicated in OA initiation and progression.^{2 14 15} Improving the integrity of subchondral bone delays the onset and progression of cartilage damage in experimental post-traumatic OA,^{16 17} and antiresorptive drugs continue to be explored as potential therapies.¹⁸ As with bone turnover, the relationship between OA and bone mineral density (BMD) is also conflicting with both increased and decreased subchondral BMD associated with OA severity and progression.^{19–21} Systemic BMD has been positively associated with thicker knee cartilage in patients with OA but also variably with progression of cartilage loss,²² although a recent meta-analysis showed no association between generalised osteoporosis and OA.²³

The relationship between OA and CKD, their ability to induce microstructural impairment in cartilage and subchondral bone, and how this modulates disease progression has not been investigated. In this study, we have developed a world first comorbid CKD-OA model in mice, and report that the metabolic changes associated with CKD affect OA subchondral bone remodelling and reduce osteophyte formation and cartilage pathology. Furthermore, we found that OA altered renal morphology and function, indicating bidirectional disease interaction.

METHODS

Animals

Male C57BL/6J (B6) mice were purchased from Australian BioResources (Sydney, Australia), housed and acclimatised for 2 weeks under a 12-hour light/dark cycle with access to standard chow and water ad libitum.

CKD and OA model

Male B6 mice aged 8–10 weeks old were randomly assigned to one of four groups: combined sham knee and sham kidney surgery (sham; n=16), 5/6 nephrectomy (5/6Nx; n=36) that subsequently had either sham (5/6Nx-sham) or destabilisation of the medial meniscus (5/6Nx-DMM) knee

surgery and DMM surgery alone (DMM; n=18), as detailed below. The 5/6Nx model is characterised by high mortality.²⁴ We altered the conventional method of nephrectomy (performed by a single surgeon (SMJ)) which significantly improved mortality to 10%. Animals were anaesthetised with isoflurane and oxygen, the left kidney exposed and exteriorised through a left flank subcostal incision, and connective tissue and adrenal gland separated by using normal saline soaked cotton bud instead of Iris scissors to prevent the injury to renal pedicle. The upper and lower renal poles were removed using Iris scissors. Haemostasis was achieved with direct pressure, the kidney returned to the abdomen and the incision closed with 5/0 monofilament nylon. At day 8 following the initial surgery, a right total nephrectomy was performed through a separate flank incision. In sham mice, the kidneys were exposed in two separate surgeries but not resected. Among 36 mice receiving 5/6Nx, 4 were euthanised (mostly on the second postoperative day) due to sickness and/or >25% body weight loss and excluded from the study. One week after right total nephrectomy, the 32 surviving 5/6Nx mice were randomly assigned to unilateral (right) sham (5/6Nx-sham, n=18) or DMM (5/6Nx-DMM, n=14) knee surgery performed by a single surgeon (CBL) as described previously.^{25–27} Briefly, the medial meniscotibial ligament was exposed through a 3mm medial parapatellar arthrotomy and infrapatellar fat pad elevation. The ligament was isolated and ruptured using dissecting forceps (ligament isolated but kept intact in sham), after which the joint was lavaged with sterile saline, and the incision closed in three layers (joint capsule and subcutaneous tissue with 8-0 Vicryl; skin with tissue glue). Mice received a single dose of buprenorphine 0.01 mg/kg at the end of all three surgeries.

Postoperatively, body weight and blood pressure (tail-cuff plethysmography; CODA machine, Kent Scientific Corporation, Torrington, CT) were measured weekly. Metabolic caging was performed 10 weeks after the first surgery for 12-hour urine collection, and at week-12 high-resolution micro-CT (μ -CT) was performed under general anaesthesia using a MILabs μ -CT-UHR (Utrecht, The Netherlands). Legs were scanned with an X-ray source of 50 kV (current: 0.24 mA, voxel size: 21 μ m and 0.4mm aluminium filter), after which mice were euthanised and blood obtained by cardiac puncture. Knee joints and kidney were processed for analysis (see below). To demonstrate model consistency, the percentage of removed kidney was calculated: % of removed kidney = weight of excised kidneys (L excised kidney + R kidney) $\times$ 100 / total wt of 2 kidneys (L excised + L remnant + R kidney).

Assessment of renal function

Urine protein and protein/creatinine ratio, blood urea and creatinine were determined using the Siemens Atellica System (at the Institute of Clinical Pathology and Medical Research, Westmead Hospital, New South Wales, Australia).

Bone morphological analysis

Following μ -CT, cross-sectional slices were generated, each scan was reconstructed using the MILabs-Rec-10.16 (Utrecht, The Netherlands), and alterations in subchondral bone were measured in sagittal, coronal and transverse sections. The morphometric parameters including fraction of bone volume (BV/TV), trabecular number (Tb.N), trabecular thickness (Tb.Th) and trabecular separation (Tb.Sp) were evaluated by an author (AD) blinded to treatment, using the built-in software—Imalytics. Bone parameters were quantified in separately defined regions of interest (ROI): whole joint—a sphere (5 mm in diameter) including distal femoral and proximal tibial metaphyseal and epiphyseal trabecular and cortical bone to enable evaluation of systemic changes; medial and lateral tibial plateau—a disc (0.5×1 mm) within the margin of tibial condyle to evaluate focal OA-related bone changes.

Renal histopathology

Kidneys were fixed in 10% neutral-buffered formalin (24 hours), embedded in paraffin, sectioned (4 μ m) and stained with H&E or picosirius red using standard methods. Slides were viewed under brightfield conditions by an independent pathologist (MS) blinded to treatment for renal tubulointerstitial lesions (TILs: dilated renal tubules, atrophic tubules and fibrosis) in five randomly selected areas as described previously with minor modifications.²⁸ Scoring of TILs was graded on a scale from 0 to 4: 0, normal; 1, mild: involvement of <25% of the cortex; 2, moderate: involvement of 25%–50% of the cortex; 3, severe: involvement of 50%–75% of the cortex and 4, extensive damage involving >75% of the cortex.

Joint histopathology

Knee joints were fixed (24 hours in 10% neutral-buffered formalin), decalcified (24 hours in 10% formic acid in 5% neutral-buffered formalin), and paraffin embedded. Sagittal 4 μ m sections every 40 μ m spanning the width of the medial tibiofemoral joint were performed. Sections were stained with toluidine blue and fast green, and digital images acquired (Nano Zoomer 210, Hamamatsu, Iwata City, Japan). Images of three serial sections encompassing the central load-bearing third of the joint in each mouse were selected and scored by a single experienced researcher (CBL) blinded to treatment as described.^{25–27} Scores were generated for: cartilage chondrocyte hypertrophy/apoptosis (0–3), proteoglycan loss (0–5) and structural damage (0–11); subchondral bone vascular invasion (0–3) and sclerosis (0–3); osteophyte maturity (0–3) and size (0–4); and synovitis (summation of synovial pannus severity (0–3) and cortical bone erosion (0–3), synoviocyte hyperplasia (0–3), subsynovial inflammatory cell infiltration (0–3), and intrasynovial exudate (0–1)). Maximum (worst) and average (from the three sections indicating spread of the lesions; all outcomes other than synovitis) scores were calculated. Cartilage changes were scored separately in the femur and tibia, followed by summation, synovitis was scored separately

in the anterior and posterior regions and summed, and bone changes were scored in the proximal tibia (as this has the most robust and consistent change).

Tartrate-resistant acid phosphatase staining

Intervening 4 μ m sections were stained in tartrate-resistant acid phosphatase (TRAP) solution (1M Na.Acetate buffer pH 5.0, 2.2 mM Naphthol AS-BI phosphate dissolved in dimethylformamide, 0.1% Sodium Nitrite) for 1 hour at 37°C as described previously.²⁹ The slides were counter-stained in 0.5% Methyl Green and mounted in DPX (Sigma-Aldrich). Images were obtained on the Aperio CS2 slide scanner (Leica Biosystems) and quantified by regions by an author (AKO), blinded to a treatment group using FIJI as described previously.³⁰

Immunohistochemistry

Immunostaining was performed as previously described.³¹ Briefly, 4 μ m sections (intervening levels between the sections used for pathology scoring) from 5 to 8 randomly selected mice from each group, were dewaxed, rehydrated and digested with 500 U/mL bovine testicular hyaluronidase (Sigma-Aldrich, St. Louis, MO) before overnight incubation with polyclonal rabbit anti-rat matrix metalloproteinase (MMP)-13 that cross reacts with mouse (LSBio, LS-B3168, 1.5 μ g/mL), in Dako antibody diluent (#S0809, Agilent Technologies) or isotype-matched IgG. For immunodetection, Dako EnVision+System HRP labelled polymer detection kit (Dako) was used with ImmPACT NovaRED Peroxidase Substrate (Vector Laboratories, Burlingame, California, USA), counterstained by Mayer's haematoxylin and Scott's bluing solution, and digital images acquired (Nano Zoomer). All samples were stained simultaneously to exclude between-run variability. The number of MMP-13 positive chondrocytes in anterior, central and posterior regions of tibial and femoral non-calcified and calcified cartilage (demarcated by tidemark) were counted (average from two independent scorers), summed and presented as either total tibial, femoral or tibiofemoral. The percentage of a standard ROI in the anterior tibial synovial fossa stained for MMP-13 was quantified using ImageJ.

Statistical analysis

Statistical analyses were performed by using GraphPad Prism software. All data are presented as mean $\pm$ SD, * p <0.05, ** p <0.01, *** p <0.001, **** p <0.0001 by Student's t-test (two groups, parametric data) or one-way analysis of variance followed by a Sidak post hoc test for multiple (n >2) group comparisons.

RESULTS

DMM induces CKD phenotypes in B6 mice

We developed an OA/CKD comorbid model as illustrated in online supplemental figure 1A. The 5/6Nx resulted in removal of 70%–80% total kidney mass (figure 1A), and significantly reduced gain of body weight in both 5/6Nx-sham and 5/6Nx-DMM mice (figure 1B). Blood pressure

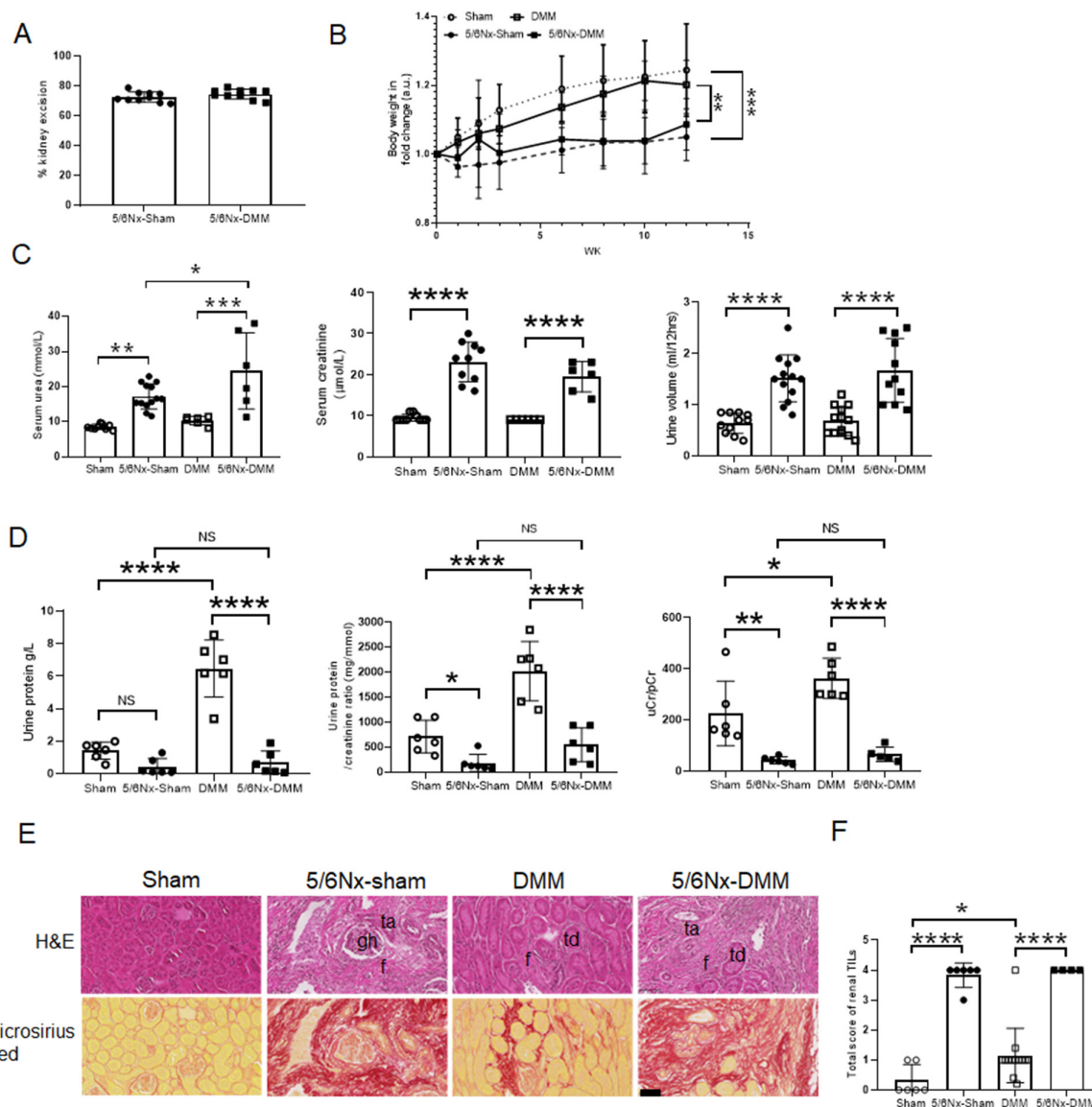


Figure 1 Analysis of chronic kidney disease (CKD) phenotypes in 5/6Nx and DMM mice. (A) Kidney excision percentage in 5/6Nx-sham (n=10) and 5/6Nx-DMM (n=10) mice. (B) Fold change of body weight of sham (n=16), 5/6Nx-sham (n=18), DMM (n=18) and 5/6Nx-DMM (n=12) mice across 12 weeks. (C) Serum urea (mmol/L), serum creatinine (μmol/L) and urine volume (mL/12 hours) in sham (n=9–11), 5/6Nx-sham (n=10–13), DMM (n=6–13) and 5/6Nx-DMM (n=6–11) mice. (D) Urine protein (g/L), urine protein/creatinine ratio (mg/mmol) and urine creatinine/plasma creatinine ratio (uCr/pCr) in sham (n=6), 5/6Nx-sham (n=6), DMM (n=6) and 5/6Nx-DMM (n=5–6). (E) Representative H&E and picrosirius stained kidney sections (Scale bar=50 μm). (F) Semiquantitative score of tubulointerstitial lesions in sham (n=6), 5/6Nx-sham (n=6), DMM (n=13) and 5/6Nx-DMM (n=4) mice. Graphs are mean±SD *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001 by one-way ANOVA with Sidak's multiple comparisons test. ANOVA, analysis of variance; DMM, destabilisation of medial meniscus; f, fibrosis; gh, glomerular hypertrophy; ta, tubular atrophy; td, tubular dilation.

and heart rate did not differ significantly between any group (online supplemental figure 1B). Renal dysfunction in 5/6Nx-sham mice was demonstrated by elevated serum urea, serum creatinine and urine volume (figure 1C), and decreased urine creatinine/plasma creatinine ratio (uCr/pCr) (figure 1D).

DMM alone increased urine protein and uCr/pCr (figure 1D), and DMM combined with 5/6Nx increased serum urea (figure 1C), showing an unanticipated impact of OA induction on kidney function. Histological examination of the left kidney revealed significantly

increased TILs in 5/6Nx-sham and 5/6Nx-DMM mice (figure 1E,F). Consistent with figure 1D, DMM alone also increased renal TILs compared with sham although significantly less than in 5/6Nx-DMM mice (figure 1D,E).

5/6Nx reduces DMM-induced histological cartilage and bone pathology

Joint histology in DMM-mice revealed typical features of OA, including non-calcified articular cartilage chondrocyte hypertrophy/apoptosis, proteoglycan loss and fibrillation/erosion, subchondral bone sclerosis with

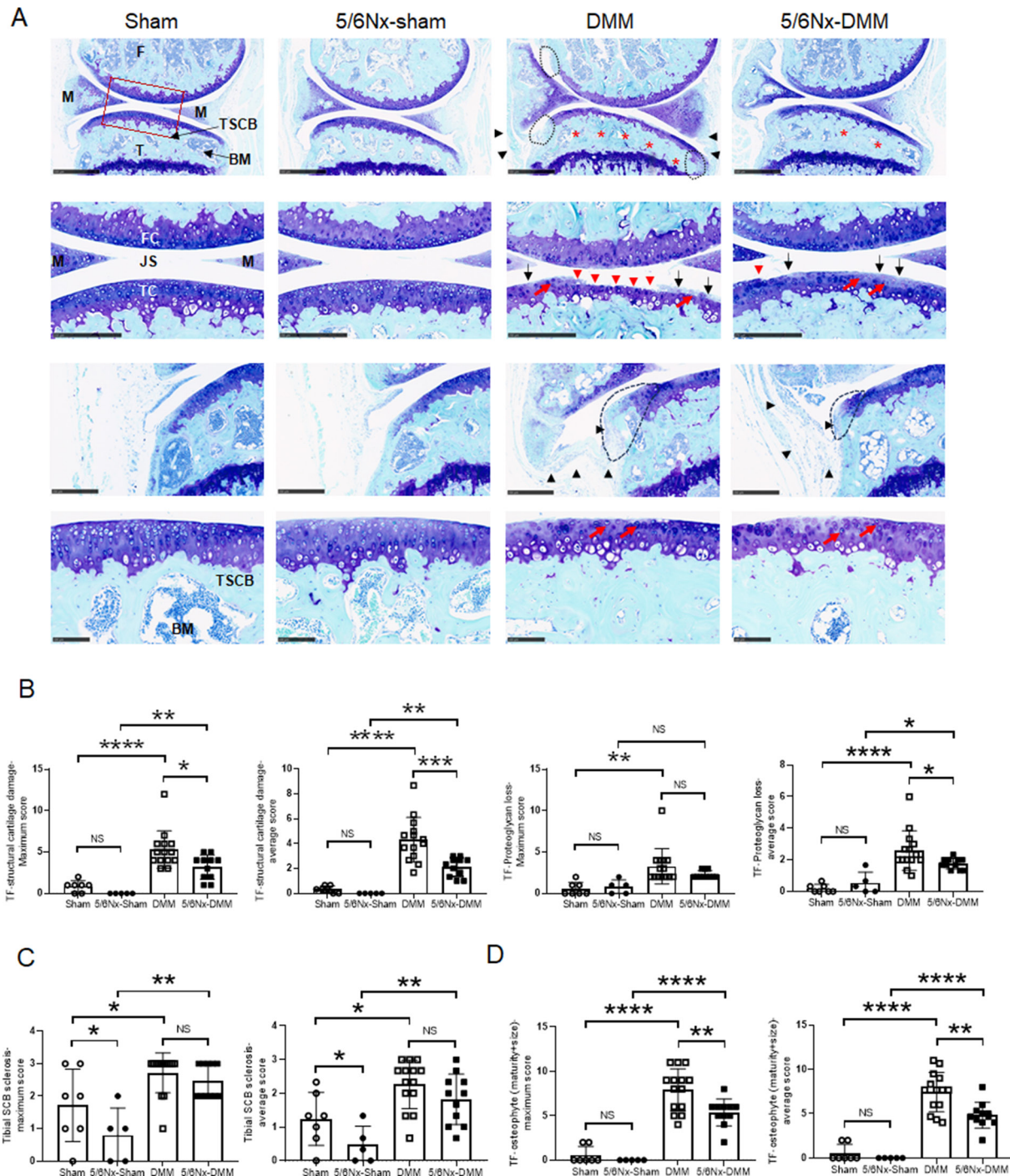


Figure 2 Histological analysis of joint pathology. Representative toluidine blue and fast green stained sections from mice 12 weeks after sham or 5/6 nephrectomy and 10 weeks after sham or DMM surgery. (A) Medial tibiofemoral compartment of joints (upper row, scale bar 500 μ m) and higher magnification images of: the central weight bearing region indicated by the red box in the upper panel (second row; scale bar 250 μ m), the anterior tibia and synovial fossa (third row; scale bar 250 μ m), and tibial articular cartilage and sub-chondral bone (forth row; scale bar 100 μ m). OA pathology in DMM joints included: non-calcified articular cartilage chondrocyte hypertrophy/apoptosis (red arrow), proteoglycan loss (reduced toluidine blue staining; black arrows), fibrillation and loss (red arrow heads), subchondral bone sclerosis with accompanying bone marrow loss (red asterisks), marginal osteophyte formation (dashed circle) and synovitis (black arrow heads). (B) Maximal and averaged tibiofemoral (TF) scores of structural cartilage damage and proteoglycan loss. (C) Maximal and averaged scores of tibial subchondral bone sclerosis. (D) Maximal and averaged scores of tibiofemoral osteophyte maturity and size. Graphs represent mean $\pm$ SD; (sham, n=7, 5/6Nx-Sham, n=5; DMM, n=14; 5/6Nx-DMM, n=11). * p <0.05, ** p <0.01, *** p <0.001, **** p <0.0001 by one-way ANOVA with Sidak's multiple comparisons test. ANOVA, analysis of variance; BM, bone marrow; DMM, destabilisation of medial meniscus; F, femur; FC, femoral cartilage; JS, joint space; M, meniscus; OA, osteoarthritis; T, tibia; TC, tibial cartilage; TSCB, tibial subchondral bone.

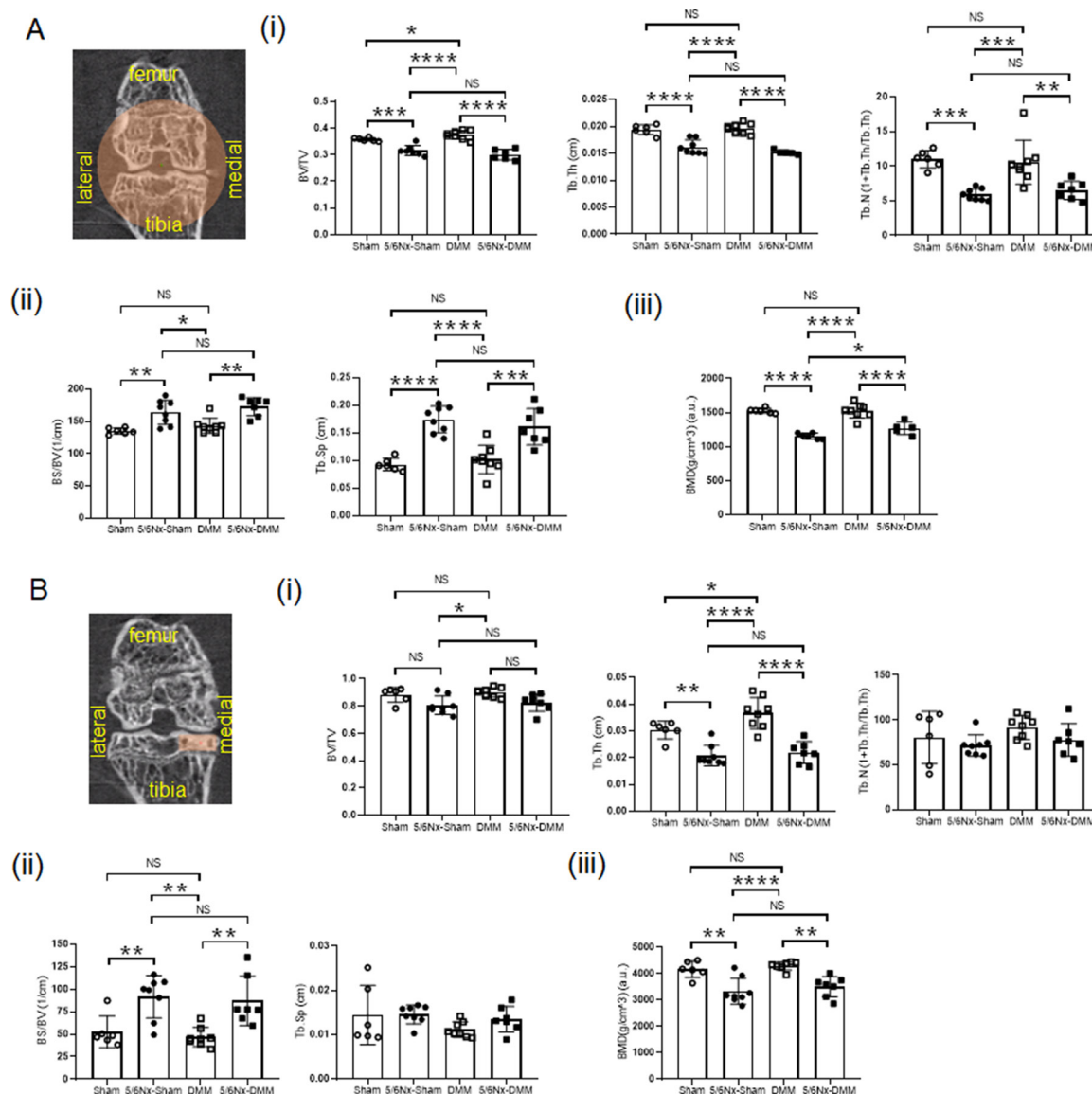


Figure 3 μ CT analysis of joint in mice at week 12 after first surgery. (A) Whole joint region of interest (ROI), diameter=5 mm; (B) medial tibial plateau ROI, diameter=(0.5×1 mm). (i) Bone volume fraction (BV/TV), trabecular thickness (Tb.Th) and trabecular number (Tb.N). (ii) Specific bone surface BS/BV and trabecular space (Tb.Sp). (iii) Bone mineral density (BMD). All data are mean±SD; (sham, n=6, 5/6Nx-Sham, n=6–87; DMM, n=8; 5/6Nx-DMM, n=5–7) * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$ by one-way ANOVA with Sidak's multiple comparisons test. ANOVA, analysis of variance; BS/BV, bone surface/bone volume; DMM, destabilisation of medial meniscus; μ CT, micro-CT.

accompanying bone marrow loss, marginal osteophyte formation, and synovitis (figure 2A). Blinded scoring revealed that maximum and average scores for all joint-wide OA-pathology features were significantly increased in DMM compared with sham-operated mice which retained normal joint histomorphology (figure 2B–D; online supplemental figure 2A–C). In 5/6Nx-sham joints, there was no increase in scores indicative of OA-based cartilage pathology (figure 2B; online supplemental figure 2A), synovitis (online supplemental figure 2B) or osteophyte/enthesophyte development (figure 2D; online supplemental figure 2C), and maximum and average subchondral bone sclerosis was significantly decreased compared with sham (figure 2C). Importantly, 5/6Nx significantly reduced DMM-induced maximal and

average articular cartilage proteoglycan loss and structural damage (figure 2B) as well as osteophyte development (figure 2D). DMM-induced subchondral bone sclerosis (figure 2C), chondrocyte hypertrophy (online supplemental figure 2A) and synovitis (online supplemental figure 2B) were not significantly changed by 5/6Nx. Overall, our data indicate that 5/6Nx reduces the severity of DMM-induced OA structural joint pathology.

5/6Nx induces loss of subchondral trabecular BV and mineral density

Analysis of μ -CT from the whole joint ROI (figure 3A) revealed that 5/6Nx induced significant loss of trabecular bone. Compared with sham, 5/6Nx-sham mice had decreased BV/TV, Tb.Th and Tb.N (figure 3Ai), with

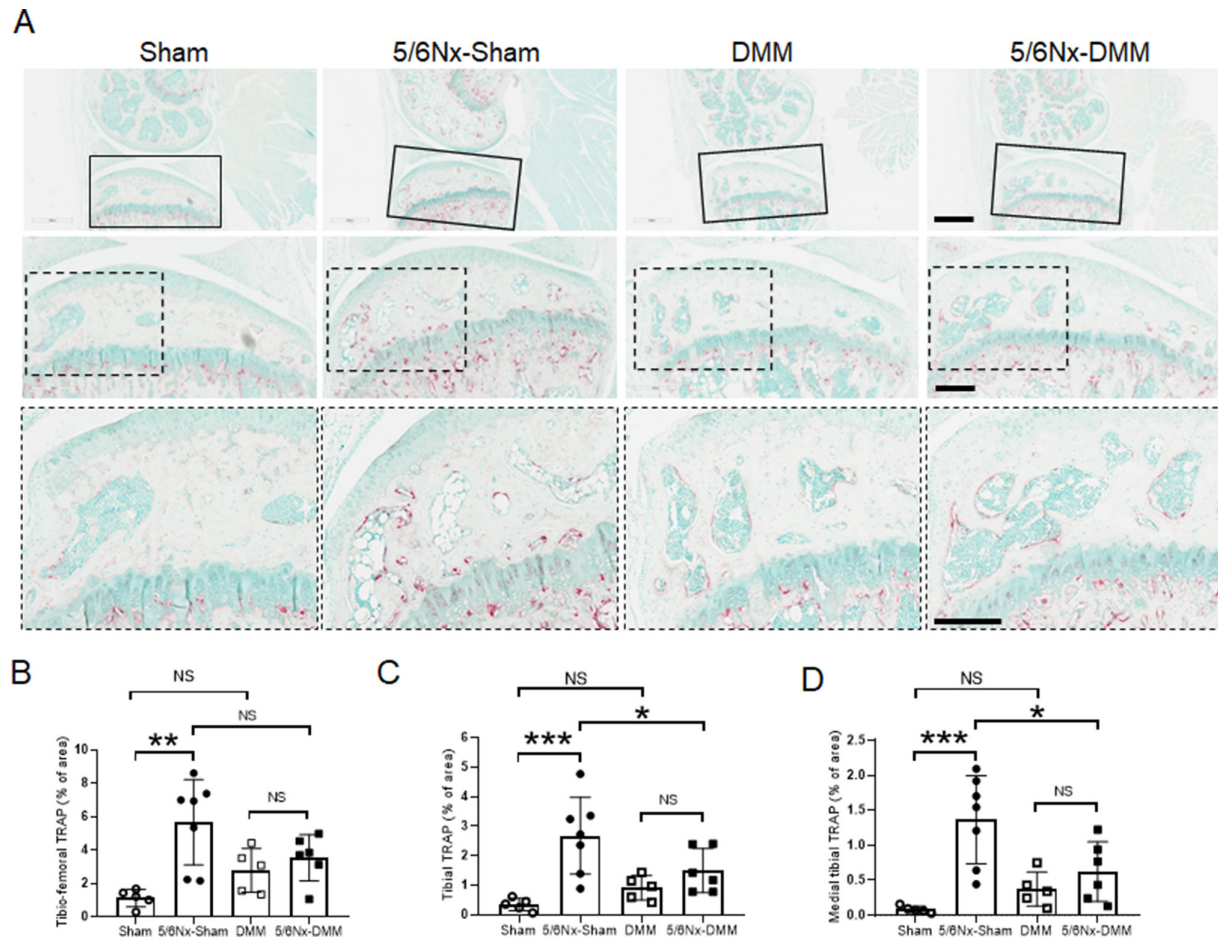


Figure 4 Tartrate-resistant acid phosphatase (TRAP) staining. (A) TRAP staining of medial tibiofemoral subchondral bone (upper panel, scale bar 600 μm), medial tibial subchondral bone (middle panel, scale bar 300 μm) and anterior medial subchondral bone (lower panel: scale bar 200 μm). Red staining indicates TRAP positive cells. Solid boxes indicate magnified standard ROI for medial tibial subchondral bone and dashed line boxes indicate magnified standard ROI for anterior medial tibial subchondral bone (see figure 2A for labelling of different joint anatomical regions and features). Semiquantitative score of TRAP staining in (B) medial tibiofemoral, (C) medial tibial and (D) anterior medial tibial subchondral bone. All data are mean±SD; (sham, n=5, 5/6Nx-Sham, n=7; DMM, n=5; 5/6Nx-DMM, n=6). *p<0.05, **p<0.01, ***p<0.001 by one-way ANOVA with Sidak's multiple comparisons test. ANOVA, analysis of variance; DMM, destabilisation of medial meniscus; ROI, regions of interest.

concomitant increases in bone surface (BS/BV) and Tb.Sp (figure 3Aii). Whole-joint bone changes induced by 5/6Nx were largely recapitulated when medial and lateral tibial plateau ROIs were evaluated separately (figure 3B and online supplemental figure 3, respectively). DMM alone significantly increased whole joint BV/TV (figure 3Ai), and medial tibial plateau Tb.Th (figure 3Bi), but did not change any measures in the lateral tibial plateau (online supplemental figure 3). There was no difference in any BV measures between 5/6Nx-sham and 5/6Nx-DMM mice in whole joint or tibial plateau ROIs (figure 3A,B and online supplemental figure 4). Thus, compared with DMM alone, 5/6Nx-DMM mice had significantly reduced whole joint BV/TV, Tb.Th and Tb.N and medial plateau Tb.Th, while whole-joint BS/BV and Tb.Sp and medial plateau Tb.Sp were significantly increased (figure 3A,B).

Similarly, BMD was significantly reduced in 5/6Nx-sham compared with sham in whole-joint and tibial plateau ROIs (figure 3Aiii,Biii and online supplemental

figure 3iii). DMM-alone had no effect on BMD in any region, and 5/6Nx-DMM mice had significantly lower BMD than DMM alone in all regions, although the reduction of whole joint BMD in 5/6Nx-DMM mice was less than 5/6Nx-sham (figure 3iii). Taken together, our data indicate 5/6Nx-induced CKD resulted in significant bone resorption, that persisted in the face of and inhibited later-stage focal DMM-induced medial tibial plateau subchondral bone formation.

DMM reduces 5/6Nx-induced subchondral osteoclasts

Osteoclast number/activity as measured by TRAP staining was increased in association with elevated subchondral bone in turnover in patients with OA³³ and early post-DMM in mice prior to later-stage bone sclerosis.¹⁴ In sham-operated mice, basal TRAP staining was present in epiphyseal and metaphyseal trabecular bone, being restricted to cells adjacent to the resorption front of the proximal tibial and distal femoral growth plates (figure 4A, online supplemental figure 4A). Epiphyseal

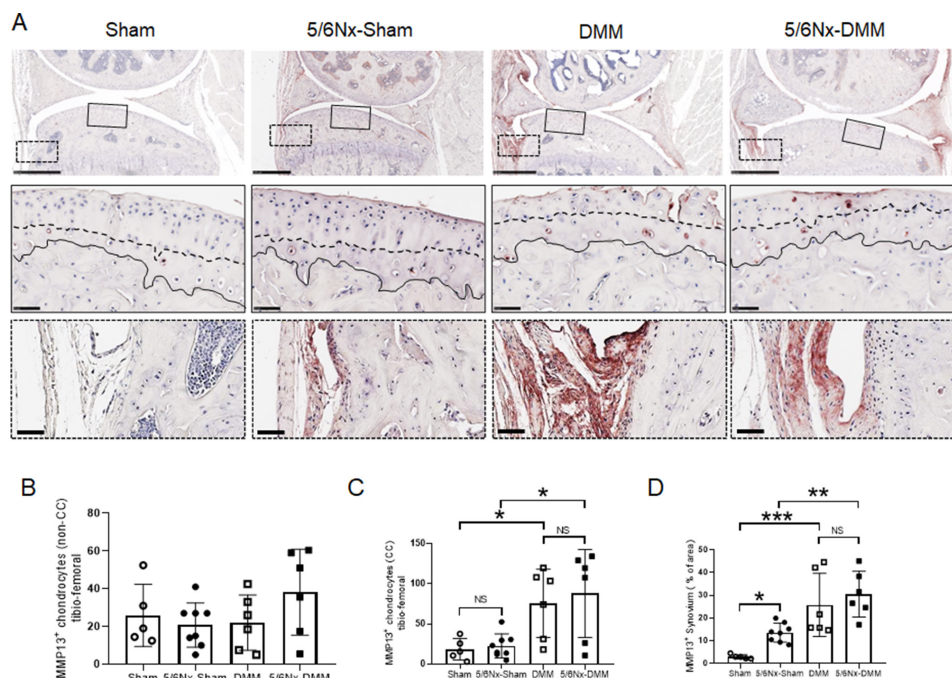


Figure 5 Matrix metalloproteinases-13 (MMP-13) in articular cartilage and synovium. (A) MMP-13 staining of medial tibiofemoral cartilage (upper panel, scale bar 500 μ m), central tibial cartilage (middle panel, scale bar 50 μ m) and anterior synovial fossa (lower panel-scale bar 50 μ m)—see figure 2A for labelling of different joint anatomical regions and features. Solid boxes indicate magnified standard ROI for central medial tibial cartilage and dashed-line boxes indicate magnified standard ROI for anterior synovial fossa. In the magnified images the dashed line demarcates the tidemark between non-calcified and calcified cartilage while the solid line indicates the osteochondral junction. Semiquantitative score of MMP-13 staining in (B) hyaline cartilage, (C) medial tibiofemoral calcified cartilage and (D) anterior synovial fossa. All data are mean $\pm$ SD; (sham, n=5, 5/6Nx-Sham, n=8; DMM, n=6; 5/6Nx-DMM, n=6). * p <0.05, ** p <0.01, *** p <0.001 by one-way ANOVA with Sidak's multiple comparisons test. ANOVA, analysis of variance; DMM, destabilisation of medial meniscus; ROI, regions of interest.

(and metaphyseal) trabecular bone TRAP staining was significantly increased in 5/6Nx-sham mice (figure 4A,B, online supplemental figure 4A,B). In DMM-alone there was no significant difference in epiphyseal TRAP-staining compared with sham-operated mice as expected at this 10-week time point. Interestingly, DMM reduced the 5/6Nx-induced tibiofemoral epiphyseal TRAP staining, this effect being most evident in the anterior medial tibial plateau (figure 4A–D, online supplemental figure 4A–D).

5/6Nx increases synovial but not cartilage MMP-13

MMPs collectively degrade all components of the extracellular matrix and play a role in resorbing both the bone and articular cartilage matrix,¹⁵ with MMP-13 in particular having a central role in both animal models²⁶ and human OA.^{33–34} In sham-operated mice, very little MMP-13 was detected, with mild positive immunostaining in scattered cells in the cartilage (particularly calcified cartilage), meniscus and synovial lining (figure 5A). While there was no change in MMP-13 in cartilage in 5/6Nx-sham mice, staining in both synovial lining and subsynovial cells was increased (figure 5A–D). In contrast, DMM increased MMP-13 in both the calcified cartilage and synovium, and this was not modified further in 5/6Nx-DMM mice (figure 5A,C and D). There was no difference between groups in MMP-13 in non-calcified cartilage (figure 5B).

DMM alters 5/6Nx-induced biochemical and bone metabolic markers in mice

The association between changes in alkaline phosphatase (ALP), calcium and phosphate homeostasis and adverse bone outcomes in CKD has been emerging.^{35–36} Consistent with this patient data, we found that serum ALP (figure 6A), phosphate (figure 6B) and calcium (figure 6C) were all significantly increased by 5/6Nx compared with sham. There was no significant effect of DMM alone on serum ALP, phosphate or calcium compared with sham, however the 5/6Nx-induced increases in serum phosphate and calcium were significantly reduced by comorbid 5/6Nx-DMM (figure 6B,C).

DISCUSSION

This is the first study to investigate the effects of comorbid induction of CKD and OA in a preclinical model, and here we report evidence of significant bidirectional disease-modifying effects. The mechanisms underpinning OA/CKD cross-talk appeared at least in part to be related to altered subchondral bone turnover. Consistent with changes seen in patients¹³ and a previous study in rats,³⁶ we now show that B6 mice are also susceptible to CKD-MBD demonstrating 5/6Nx-induced decreased subchondral BV and BMD through increased osteoclast formation/activity with associated increases in serum

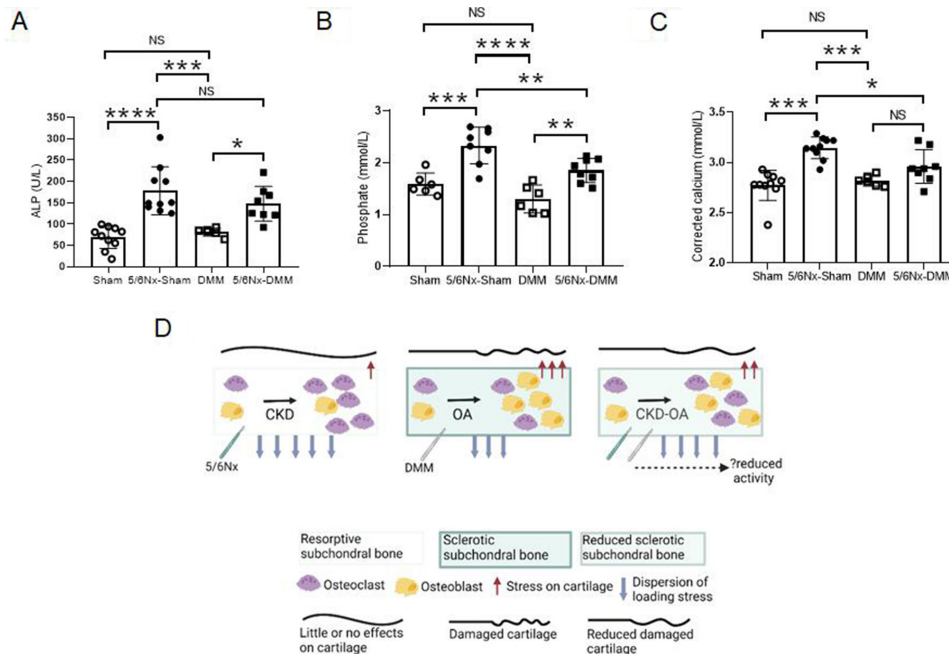


Figure 6 Biochemical analysis of serum markers. (A) Alkaline phosphatase (U/L), (B) phosphate (mmol/L) and (C) corrected calcium levels (mmol/L). All data are mean $\pm$ SD; (sham, n=6–10, 5/6Nx-Sham, n=8–10; DMM, n=6; 5/6Nx-DMM, n=8). *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001 by one-way ANOVA with Sidak's multiple comparisons test. (D) A schematic model of osteoarthritis (OA) and chronic kidney disease (CKD). Dominant bone formation in subchondral bone in OA altered by CKD induced resorptive subchondral bone might reduce the loading stress to cartilage. ANOVA, analysis of variance; DMM, destabilisation of medial meniscus.

calcium and phosphate. Increased differentiation of circulating inflammatory monocytes into bone-resorptive osteoclasts is part of the CKD-MBD syndrome associated with secondary hyperparathyroidism and vascular pathology.^{37–39} While partially offset by later-stage OA-induced focal bone formation, the persistence of 5/6Nx-induced subchondral bone turnover and reduced BMD may ameliorate excessive mechanical loading on the overlying cartilage associated with OA-induced subchondral sclerosis as well as abnormal biological cross-talk from diseased subchondral osteocytes.¹⁵

In addition to subchondral bone remodelling, osteoclasts are critical in regulating calcified cartilage turnover in endochondral ossification. While long-bone growth had ceased in the mice in the current study the CKD-induced osteoclast activity may contribute to the reduction in osteophyte size. Osteophytes form as a result of inflammatory and biomechanical signals which drive chondroid differentiation of stem cells in the joint margin and outgrowth of cartilage precursors (chondrocytes) that undergo endochondral ossification.^{25, 40} As well as decreasing chondrocyte growth and maturation,⁹ CKD-induced synovial MMP-13 and osteoclast formation/activity may accelerate chondrocyte matrix degradation and new bone turnover, thereby reducing osteophyte development. Our data in mice are consistent with the more erosive OA phenotype with limited osteophyte formation in patients with CKD receiving dialysis.¹¹

Cartilage degradation and osteophyte formation could also be reduced as a result of physical inactivity driven

by 5/6Nx-induced ill health and less mechanical signaling, as reported with joint unloading following DMM.⁴¹ We did not measure activity levels but the reduced body weight gain in both 5/6Nx-sham and 5/6Nx-DMM groups (compared with sham kidney-surgery or DMM alone) is consistent with CKD-specific systemic disease. Metabolic dysfunction can result in systemic proinflammatory state in patients with CKD^{42, 43} and increased synovial MMP-13 in 5/6Nx mice would be in keeping with this. CKD-priming of a procatabolic joint environment could also promote cartilage degradation and OA risk with time and/or additional stimuli (eg, obesity, age, joint injury).^{5, 44} However, 5/6Nx did not increase articular cartilage chondrocyte MMP-13, suggesting not only that the inflammatory/procatabolic pathways activated by CKD may be tissue-specific but that chondrocyte (rather than synovial) MMP-13 may be more important for cartilage degradation. DMM increased both synovial and chondrocyte MMP-13, the latter predominantly in calcified cartilage, likely as a result of both inflammatory and biomechanical signals.^{15, 41} That 5/6Nx did not augment chondrocyte MMP-13 enabled CKD-induced subchondral bone remodelling to be chondro-protective in DMM joints.

In addition to 5/6Nx modulating DMM-induced OA, there was also modification of 5/6Nx-induced metabolic abnormalities by OA, and surprisingly, a direct effect of DMM on renal pathology. Decreased serum calcium and phosphate in 5/6Nx-DMM mice likely reflect increased bone formation and reduced bone turnover at this stage of

OA, counteracting the 5/6Nx-induced osteoclast activity. It would be interesting to determine in future studies whether serum calcium and phosphate may have actually been elevated earlier post-DMM when subchondral bone resorption is increased.²⁵ How 5/6Nx-DMM increased serum urea is unclear but may be indicative of increased tissue catabolism both locally in the affected joint and in more distant muscle groups, for example, disuse atrophy.⁴⁵ Unexpectedly, DMM alone induced renal dysfunction as measured by proteinuria, altered uCr/pCr and increased fibrosis. The renopathic effects of DMM occurred in the absence of confounding risk factors such as non-steroidal anti-inflammatory drug administration.⁴⁶ The profibrotic effect of DMM was substantially less than that of 5/6Nx which alone maximised kidney scores, masking any potential effects of comorbid OA at 12 weeks. Future studies should examine earlier time points to determine whether DMM/OA alters the time course of 5/6Nx-induced fibrosis. The mechanisms driving the OA/CKD link is unclear, and future studies will investigate the temporal regulation and potential role in OA-associated renal pathology of factors such as changes in circulating inflammatory cytokines⁴⁷ and/or release of bioactive micro-RNAs from diseased joints.^{48–50}

Our observation of a bidirectional OA/CKD interaction in mice is consistent with longitudinal patient data but with an important difference: in humans each disease increases incident risk of the other,⁵ while in our mouse model OA increased renal disease but CKD reduced OA severity. This discordance may be related to timing and the duration of disease interaction (10 weeks in mice vs years in patients). A previous study in rats found cartilage proteoglycan content was not affected until 16 weeks post-5/6Nx with increased cartilage degradation measured at 44 weeks.³⁹ These data suggest the duration of comorbid disease interaction may modify effects, and future studies will examine longer time points to establish if OA protection seen at 10 weeks in the current study is lost with time. Other variables that could contribute to differences between our model and the human studies are age (adolescent mice vs aged patients), sex (male mice vs mixed sex patients) and lack of additional comorbidities in our mice (eg, obesity, diabetes) that affect joint structure and function.^{5 44} These confounding effects should be explored in the future.

In conclusion, this study has established a novel preclinical comorbid model for CKD-OA and identified significant bidirectional disease-modifying effects. Cartilage sparing appeared to be associated with CKD-induced remodelling and microarchitectural changes in the subchondral bone, potentially by altering loading stress transmission in the functional subchondral bone articular cartilage unit (figure 6D). Unexpectedly, DMM modified 5/6Nx-CKD metabolic changes, and importantly on its own induced kidney fibrosis and proteinuria. CKD presents major issues for managing OA pain in patients

with available analgesics increasing the risk of CKD progression. The new preclinical model we have described offers an exciting opportunity to explore the molecular pathways involved in OA-CKD cross-talk, and to define new therapeutic approaches by driving drug development.

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